# Network Threats and Mitigation & Physical Security and Risk

**Task 1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).**

**Answer:**

| No. | Threat | What it is | Real-world example (app / service) | How to reduce it |
|---|---|---|---|---|
| 1 | Phishing | Fake messages or links that trick a user into giving passwords, OTPs or card details. | Paytm: an SMS says "Your Paytm KYC has expired, click this link to update", and the link opens a fake page that asks for the OTP and UPI PIN. | Never click unknown links; Paytm never asks for PIN or OTP; report the message. |
| 2 | DDoS (Distributed Denial of Service) | Attackers send a huge amount of fake traffic from many devices, so the real users cannot use the service. | A ticket-booking or food-delivery app (like Zomato or IRCTC) becomes very slow or shows errors at peak time because bots flood its servers with fake requests. | Traffic filtering, rate limiting and DDoS protection services (CDN / cloud firewall). |
| 3 | Man-in-the-Middle (MITM) | An attacker secretly sits between the user and the website and reads or changes the data. | Instagram or YouTube login on a free public Wi-Fi at a cafe: a fake hotspot with a similar name can capture the traffic. | Use HTTPS only, avoid public Wi-Fi for logins, and use a VPN. |
| 4 | Malware | Harmful software (virus, spyware, trojan) that damages the device or steals data. | WhatsApp: a forwarded message offers a "free IPL live app" APK; after installing it, the app steals contacts and messages. | Install apps only from Play Store / App Store and keep antivirus and updates on. |
| 5 | Password attacks (brute force / credential stuffing) | The attacker guesses passwords or tries leaked passwords on other sites. | Instagram: someone tries thousands of passwords, or a password leaked from another site, to log in to an account that uses the same password. | Use long unique passwords, two-factor authentication and account lock after failed attempts. |

**Task 2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.**

**Answer:**

The script has a preset PIN ("7391"). It tries every number from 0000 to 9999, one by one, and counts each try. When the guess matches the preset PIN, it prints the PIN and the number of attempts. A 4-digit PIN has only 10,000 combinations, so a computer can try all of them in a fraction of a second. This is why PINs need an attempt limit and lock.

**Code (brute_force.py):**

```
# Brute-force simulation: guess a 4-digit PIN by trying every combination
```

```
SECRET_PIN = "7391"          # preset value (the PIN we are trying to guess)


def brute_force(secret):
    attempts = 0
    for number in range(10000):              # 0000 to 9999
        guess = str(number).zfill(4)         # keep 4 digits, e.g. 7 -> "0007"
        attempts += 1
        if guess == secret:
            return guess, attempts
    return None, attempts


found_pin, total_attempts = brute_force(SECRET_PIN)

if found_pin:
    print(f"PIN found: {found_pin}")
    print(f"Number of attempts: {total_attempts}")
else:
    print("PIN not found")
```

**Output:**

```
PIN found: 7391
Number of attempts: 7392
```

Explanation: The PIN 7391 is the 7392nd combination (counting from 0000), so the script needed 7392 attempts. If the preset PIN is changed, the number of attempts changes too.

### Task 3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

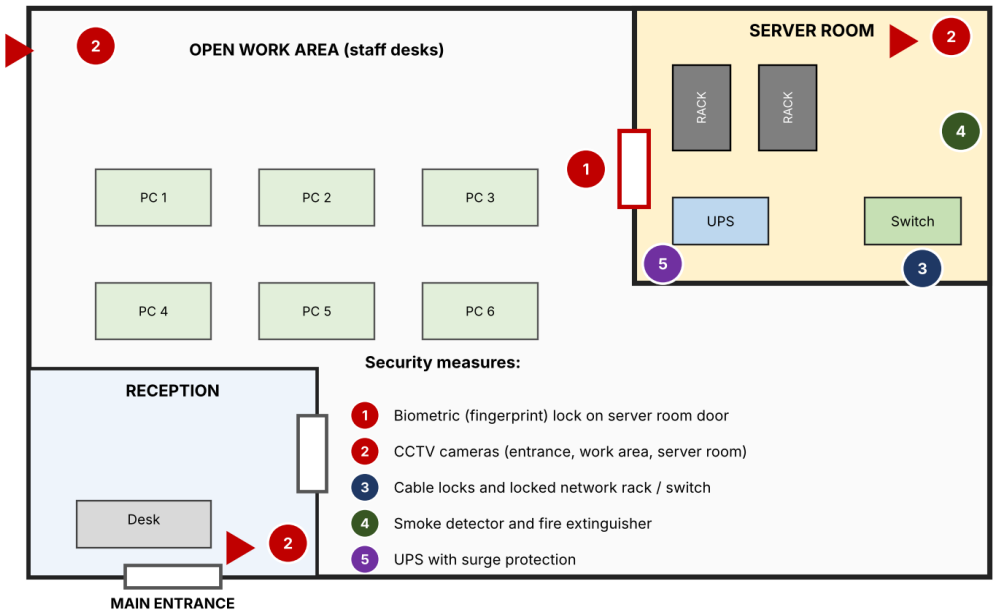
**Answer:**

| No. | Physical security risk | What can happen | Mitigation step |
|---|---|---|---|
| 1 | Unauthorized physical access / theft | A stranger or an untrained person enters the room, steals a hard disk or server, or plugs in a device to copy data. | Biometric or smart-card door lock with an entry log and CCTV at the door. |
| 2 | Fire | A short circuit or overheating starts a fire and burns servers, which stops the app and destroys data. | Install smoke detectors and a gas-based fire suppression system (not water) with extinguishers. |
| 3 | Power failure or power surge | A sudden power cut or voltage spike shuts the servers down or damages hardware and corrupts data. | Use UPS with surge protection and a backup diesel generator. |
| 4 | Overheating / water leakage / flooding | If the cooling (AC) fails, servers overheat and shut down. A pipe leak or flood can damage equipment. | Keep a backup AC system with a temperature alarm, and place racks above floor level with water-leak sensors. |

### Task 4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

**Answer:**

**Floor Plan: Small Office with Network Equipment Security**




Security measures marked on the plan:

1. Biometric (fingerprint) lock on the server room door, so only authorized staff can enter.
2. CCTV cameras at the main entrance, in the work area and inside the server room to record who goes where.
3. Cable locks and a locked network rack, so the switch, router and cables cannot be stolen or unplugged.
4. Smoke detector and fire extinguisher in the server room.
5. UPS with surge protection to protect the equipment from power cuts and spikes.

## Task 5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

**Answer:**

Checklist for a college computer lab (reviewed and edited for a lab with student PCs, a lab server, a switch and a Wi-Fi router):

| No. | Action for the network admin | Covers | College lab specific detail |
|---|---|---|---|
| 1 | Keep the operating system, browsers and antivirus on all PCs updated and use a restore tool (like Deep Freeze). | Network threat (malware) | Students install random software, so reset the lab PCs to a clean state after every session. |
| 2 | Turn on the firewall on the lab router and on every PC, and block ports that are not needed. | Network threat (attacks) | Block file-sharing, torrent and gaming ports; allow only the websites and ports needed for classes. |

| No. | Action for the network admin | Covers | College lab specific detail |
|---|---|---|---|
| 3 | Use strong passwords and separate accounts; change all default passwords on the router, switch and Wi-Fi. | Network threat (password attacks) | Students use limited accounts; only the admin account has install rights; no shared admin password. |
| 4 | Secure the Wi-Fi and separate the networks (WPA2/WPA3, different VLAN for students). | Network threat (MITM, unauthorized access) | Student and staff networks are kept separate, and the lab server is not reachable from student Wi-Fi. |
| 5 | Lock the network equipment and cables in a locked rack, and use cable locks on PCs. | Physical security | The switch and router are kept in a locked cabinet; only the lab in-charge has the key; USB ports can be disabled. |
| 6 | Control entry with a lab register, ID card or biometric, and install CCTV. | Physical security | Students enter only during scheduled hours with a log; the lab is locked after class; CCTV covers the door and the server. |
| 7 | Take regular backups, keep a UPS and fire extinguisher, and run a short awareness session. | Both network and physical | Daily backup of the lab server; UPS for the server and switch; teach students about phishing and safe use of lab PCs. |